often referred to as *subject-specific* models, and marginal models are called *population-averaged* models. In Subsection 2.3.1, we specify the marginal model in general, and in Subsection 2.3.2, we present the marginal model implied by an LMM.

2.3.1 Specification of the Marginal Model

The general matrix specification of the marginal model for subject i is

$$\mathbf{Y}_i = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{\varepsilon}_i^* \tag{2.6}$$

where

$$\boldsymbol{\varepsilon}_i^* \sim \mathcal{N}(0, \boldsymbol{V}_i^*)$$

The $n_i \times p$ design matrix $\boldsymbol{X}_i$ is constructed the same way as in an LMM. Similarly, $\boldsymbol{\beta}$ is a vector of fixed effects. The vector $\boldsymbol{\varepsilon}_i^*$ represents a vector of **marginal residual errors.** Elements in the $n_i \times n_i$ marginal covariance matrix $\boldsymbol{V}_i^*$ are usually defined by a small set of covariance parameters, which we denote as $\boldsymbol{\theta}^*$. All structures used for the $\boldsymbol{R}_i$ matrix in LMMs (described in Subsection 2.2.2.2) can be used to specify a structure for $\boldsymbol{V}_i^*$. Other structures for $\boldsymbol{V}_i^*$, such as those shown in Subsection 2.3.2, are also allowed.

Note that the entire random part of the marginal model is described in terms of the marginal residual errors $\boldsymbol{\varepsilon}_i^*$ only. In contrast to the LMM, the marginal model does not involve the random effects, $\mathbf{u}_i$, so inferences cannot be made about them and consequently this model is not a *mixed* model.

Software Note: Several software procedures designed for fitting LMMs, including the procedures in SAS, SPSS, R, and Stata, also allow users to specify a marginal model directly. The most natural way to specify selected marginal models in these procedures is to make sure that random effects are not included in the model, and then specify an appropriate covariance structure for the $\boldsymbol{R}_i$ matrix, which in the context of the marginal model will be used for $\boldsymbol{V}_i^*$. A marginal model of this form is **not** an LMM, because no random effects are included in the model. This type of model cannot be specified using the HLM software, because HLM generally requires the specification of at least one set of random effects (e.g., a random intercept). Examples of fitting a marginal model by omitting random effects and using an appropriate $\boldsymbol{R}_i$ matrix are given in alternative analyses of the Rat Brain data at the end of Chapter 5, and the Autism data at the end of Chapter 6.

2.3.2 The Marginal Model Implied by an LMM

The LMM introduced in (2.2) implies the following marginal linear model:

$$\mathbf{Y}_i = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{\varepsilon}_i^* \tag{2.7}$$

where

$$\boldsymbol{\varepsilon}_i^* \sim \mathcal{N}(0, \boldsymbol{V}_i)$$

and the covariance matrix, $\boldsymbol{V}_i$, is defined as

$$\boldsymbol{V}_i = \boldsymbol{Z}_i\boldsymbol{D}\boldsymbol{Z}_i' + \boldsymbol{R}_i.$$

A few observations are in order. First, the implied marginal model is an example of the marginal model defined in Subsection 2.3.1. Second, the LMM in (2.2) and the corresponding implied marginal model involve the same set of covariance parameters $\boldsymbol{\theta}$ (i.e., the $\boldsymbol{\theta}_D$ and $\boldsymbol{\theta}_R$ vectors combined). The important difference is that there are more restrictions imposed on the covariance parameter space in the LMM than in the implied marginal model. For example, the diagonal elements (i.e., variances) in the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices of LMMs are required to be positive. This requirement is not needed in the implied marginal model. More generally, the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices in LMMs have to be positive-definite, whereas the only requirement in the implied marginal model is that the $\boldsymbol{V}_i$ matrix be positive-definite. Third, interpretation of the covariance parameters in a marginal model is different from that in an LMM, because inferences about random effects are no longer valid.

The concept of the implied marginal model is important for at least two reasons. First, estimation of fixed-effect and covariance parameters in the LMM (see Subsection 2.4.1.2) is carried out in the framework of the implied marginal model. Second, in the case in which a software procedure produces a nonpositive definite (i.e., invalid) estimate of the $\boldsymbol{D}$ matrix in an LMM, we may be able to fit the implied marginal model, which has fewer restrictions. Consequently, we may be able to diagnose problems with nonpositive definiteness of the $\boldsymbol{D}$ matrix or, even better, we may be able to answer some relevant research questions in the context of the implied marginal model.

The implied marginal model defines the marginal distribution of the $\mathbf{Y}_i$ vector:

$$\mathbf{Y}_i \sim \mathcal{N}(\boldsymbol{X}_i\boldsymbol{\beta},\ \boldsymbol{Z}_i\boldsymbol{D}\boldsymbol{Z}_i' + \boldsymbol{R}_i) \tag{2.8}$$

The marginal mean (or expected value) and the marginal covariance matrix of the vector $\mathbf{Y}_i$ are equal to

$$E(\mathbf{Y}_i) = \boldsymbol{X}_i\boldsymbol{\beta} \tag{2.9}$$

and

$$Var(\mathbf{Y}_i) = \boldsymbol{V}_i = \boldsymbol{Z}_i\boldsymbol{D}\boldsymbol{Z}_i' + \boldsymbol{R}_i.$$

The off-diagonal elements in the $n_i \times n_i$ matrix $\boldsymbol{V}_i$ represent the marginal covariances of the $\mathbf{Y}_i$ vector. These covariances are in general different from zero, which means that in the case of a longitudinal data set, repeated observations on a given individual i are correlated. We present an example of calculating the $\boldsymbol{V}_i$ matrix for the marginal model implied by an LMM fitted to the Rat Brain data (Chapter 5) in Appendix B. The marginal distribution specified in (2.8), with mean and variance defined in (2.9), is a focal point of the likelihood estimation in LMMs outlined in the next section.

Software Note: The software discussed in this book is primarily designed to fit LMMs. In some cases, we may be interested in fitting the marginal model implied by a given LMM using this software:

1. For some fairly simple LMMs, it is possible to specify the implied marginal model directly using the software procedures in SAS, SPSS, R, and Stata, as described in Subsection 2.3.1. As an example, consider an LMM with random intercepts and constant residual error variance. The $\boldsymbol{V}_i$ matrix for the marginal model implied by this LMM has a compound symmetry structure (see Appendix B), which can be specified by omitting the random intercepts from the model and choosing a compound symmetry structure for the $\boldsymbol{R}_i$ matrix.

2. Another very general method available in the LMM software procedures is to "emulate" fitting the implied marginal model by fitting the LMM itself. By emulation, we mean using the same syntax as for an LMM, i.e., including specification of random effects, but interpreting estimates and other results as if they were obtained for the marginal model. In this approach, we simply take advantage of the fact estimation of the LMM and of the implied marginal model are performed using the same algorithm (see Section 2.4).

3. Compared to LMMs, the general emulation approach outlined in item 2 has some limitations related to less restrictive constraints in the implied marginal. In most software procedures that fit LMMs, it is difficult to relax the positive definiteness constraints on the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices as required by the implied marginal model. The nobound option in SAS `proc mixed` and `proc glimmix` is the only exception among the software procedures discussed in this book that allows users to remove the positive definiteness constraints on the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices and allows user-defined constraints to be imposed on the covariance parameters in the $\boldsymbol{\theta}_D$ and $\boldsymbol{\theta}_R$ vectors. An example of using the nobound option to relax the constraints on covariance parameters applicable to the fitted linear mixed model is given in Subsection 6.4.1.

2.4 Estimation in LMMs

In the LMM, we estimate the fixed-effect parameters, $\boldsymbol{\beta}$, and the covariance parameters, $\boldsymbol{\theta}$ (i.e., $\boldsymbol{\theta}_D$ and $\boldsymbol{\theta}_R$ for the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices, respectively). In this section, we discuss maximum likelihood (ML) and restricted maximum likelihood (REML) estimation, which are methods commonly used to estimate these parameters.

2.4.1 Maximum Likelihood (ML) Estimation

In general, **maximum likelihood (ML) estimation** is a method of obtaining estimates of unknown parameters by optimizing a **likelihood function**. To apply ML estimation, we first construct the likelihood as a function of the parameters in the specified model, based on distributional assumptions. The **maximum likelihood estimates (MLEs)** of the parameters are the values of the arguments that maximize the likelihood function (i.e., the values of the parameters that make the observed values of the dependent variable most likely, given the distributional assumptions). See Casella & Berger (2002) for an in-depth discussion of ML estimation.

In the context of the LMM, we construct the likelihood function of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ by referring to the marginal distribution of the dependent variable $\mathbf{Y}_i$ defined in (2.8). The corresponding **multivariate normal probability density function**, $f(\mathbf{Y}_i|\boldsymbol{\beta},\boldsymbol{\theta})$, is:

$$f(\mathbf{Y}_i|\boldsymbol{\beta},\boldsymbol{\theta}) = (2\pi)^{\frac{-n_i}{2}} \det(\boldsymbol{V}_i)^{\frac{-1}{2}} \exp(-0.5 \times (\mathbf{Y}_i - \boldsymbol{X}_i\boldsymbol{\beta})'\boldsymbol{V}_i^{-1}(\mathbf{Y}_i - \boldsymbol{X}_i\boldsymbol{\beta})) \tag{2.10}$$

where det refers to the determinant. Recall that the elements of the $\boldsymbol{V}_i$ matrix are functions of the covariance parameters in $\boldsymbol{\theta}$.

Based on the probability density function (pdf) defined in (2.10), and given the observed data $\mathbf{Y}_i = \mathbf{y}_i$, the likelihood function contribution for the i-th subject is defined as follows:

$$L_i(\boldsymbol{\beta}, \boldsymbol{\theta}; \mathbf{y}_i) = (2\pi)^{\frac{-n_i}{2}} \det(\boldsymbol{V}_i)^{\frac{-1}{2}} \exp(-0.5 \times (\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta})'\boldsymbol{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta})) \tag{2.11}$$

We write the **likelihood function**, $L(\boldsymbol{\beta}, \boldsymbol{\theta})$, as the product of the m independent contributions defined in (2.11) for the individuals ($i = 1, \ldots, m$):

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}) = \prod_i L_i(\boldsymbol{\beta}, \boldsymbol{\theta}; \mathbf{y}_i) = \prod_i (2\pi)^{\frac{-n_i}{2}} \det(\boldsymbol{V}_i)^{\frac{-1}{2}} \exp(-0.5 \times (\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta})'\boldsymbol{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta})) \tag{2.12}$$

The corresponding **log-likelihood function**, $\ell(\boldsymbol{\beta}, \boldsymbol{\theta})$, is defined as

$$\begin{aligned} \ell(\boldsymbol{\beta}, \boldsymbol{\theta}) &= \log L(\boldsymbol{\beta}, \boldsymbol{\theta}) = -0.5n \times \log(2\pi) - 0.5 \times \sum_i \log(\det(\boldsymbol{V}_i)) \\ &\quad -0.5 \times \sum_i (\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta})'\boldsymbol{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta}) \end{aligned} \tag{2.13}$$

where n ($= \sum_i n_i$) is the number of observations (rows) in the data set, and "log" refers to the natural logarithm.

Although it is often possible to find estimates of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ simultaneously, by optimization of $\ell(\boldsymbol{\beta}, \boldsymbol{\theta})$ with respect to both $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$, many computational algorithms simplify the optimization by **profiling out** the $\boldsymbol{\beta}$ parameters from $\ell(\boldsymbol{\beta}, \boldsymbol{\theta})$, as shown in Subsections 2.4.1.1 and 2.4.1.2.

2.4.1.1 Special Case: Assume $\boldsymbol{\theta}$ Is Known

In this section, we consider a special case of ML estimation for LMMs, in which we assume that $\boldsymbol{\theta}$, and as a result the matrix $\boldsymbol{V}_i$, are known. Although this situation does not occur in practice, it has important computational implications, so we present it separately.

Because we assume that $\boldsymbol{\theta}$ is known, the only parameters that we estimate are the fixed effects, $\boldsymbol{\beta}$. The log-likelihood function, $\ell(\boldsymbol{\beta}, \boldsymbol{\theta})$, thus becomes a function of $\boldsymbol{\beta}$ only, and its optimization is equivalent to finding a minimum of an objective function $\mathrm{q}(\boldsymbol{\beta})$, defined by the last term in (2.13):

$$\mathrm{q}(\boldsymbol{\beta}) = 0.5 \times \sum_i (\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta})'\boldsymbol{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{X}_i\boldsymbol{\beta}) \tag{2.14}$$

This function looks very much like the matrix formula for the sum of squared errors that is minimized in the standard linear model, but with the addition of the nondiagonal "weighting" matrix $\boldsymbol{V}_i^{-1}$.

Note that optimization of $\mathrm{q}(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ can be carried out by applying the method of **generalized least squares** (GLS). The optimal value of $\boldsymbol{\beta}$ can be obtained analytically:

$$\widehat{\boldsymbol{\beta}} = (\sum_i \boldsymbol{X}_i'\boldsymbol{V}_i^{-1}\boldsymbol{X}_i)^{-1} \sum_i \boldsymbol{X}_i'\boldsymbol{V}_i^{-1}\mathbf{y}_i \tag{2.15}$$

The estimate $\widehat{\boldsymbol{\beta}}$ has the desirable statistical property of being the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$.

The closed-form formula in (2.15) also defines a functional relationship between the covariance parameters, $\boldsymbol{\theta}$, and the value of $\boldsymbol{\beta}$ that maximizes $\ell(\boldsymbol{\beta}, \boldsymbol{\theta})$. We use this relationship in the next section to **profile out** the fixed-effect parameters, $\boldsymbol{\beta}$, from the log-likelihood, and make it strictly a function of $\boldsymbol{\theta}$.

2.4.1.2 General Case: Assume $\boldsymbol{\theta}$ Is Unknown

In this section, we consider ML estimation of the covariance parameters, $\boldsymbol{\theta}$, and the fixed effects, $\boldsymbol{\beta}$, assuming $\boldsymbol{\theta}$ is unknown.

First, to obtain estimates for the covariance parameters in $\boldsymbol{\theta}$, we construct a **profile log-likelihood function** $\ell_{ML}(\boldsymbol{\theta})$. The function $\ell_{ML}(\boldsymbol{\theta})$ is derived from $\ell(\boldsymbol{\beta}, \boldsymbol{\theta})$ by replacing the $\boldsymbol{\beta}$ parameters with the expression defining $\widehat{\boldsymbol{\beta}}$ in (2.15). The resulting function is

$$\ell_{ML}(\boldsymbol{\theta}) = -0.5n \times \ln(2\pi) - 0.5 \times \sum_i \ln(\det(\boldsymbol{V}_i)) - 0.5 \times \sum_i \mathbf{r}_i' \boldsymbol{V}_i^{-1} \mathbf{r}_i \tag{2.16}$$

where

$$\mathbf{r}_i = \mathbf{y}_i - \boldsymbol{X}_i((\sum_i \boldsymbol{X}_i' \boldsymbol{V}_i^{-1} \boldsymbol{X}_i)^{-1} \sum_i \boldsymbol{X}_i' \boldsymbol{V}_i^{-1} \mathbf{y}_i) \tag{2.17}$$

In general, maximization of $\ell_{ML}(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$ is an example of a nonlinear optimization, with inequality constraints imposed on $\boldsymbol{\theta}$ so that positive definiteness requirements on the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices are satisfied. There is no closed-form solution for the optimal $\boldsymbol{\theta}$, so the estimate of $\boldsymbol{\theta}$ is obtained by performing computational iterations until convergence (see Subsection 2.5.1).

After the ML estimates of the covariance parameters in $\boldsymbol{\theta}$ (and consequently, estimates of the variances and covariances in $\boldsymbol{D}$ and $\boldsymbol{R}_i$) are obtained through an iterative computational process, we are ready to calculate $\widehat{\boldsymbol{\beta}}$. This can be done without an iterative process, using (2.18) and (2.19). First, we replace the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices in (2.9) by their ML estimates, $\widehat{\boldsymbol{D}}$ and $\widehat{\boldsymbol{R}}_i$, to calculate $\widehat{\boldsymbol{V}}_i$, an estimate of $\boldsymbol{V}_i$:

$$\widehat{\boldsymbol{V}}_i = \boldsymbol{Z}_i \widehat{\boldsymbol{D}} \boldsymbol{Z}_i' + \widehat{\boldsymbol{R}}_i \tag{2.18}$$

Then, we use the generalized least-squares formula, (2.15), for $\widehat{\boldsymbol{\beta}}$, with $\boldsymbol{V}_i$ replaced by its estimate defined in (2.18) to obtain $\widehat{\boldsymbol{\beta}}$:

$$\widehat{\boldsymbol{\beta}} = (\sum_i \boldsymbol{X}_i' \widehat{\boldsymbol{V}}_i^{-1} \boldsymbol{X}_i)^{-1} \sum_i \boldsymbol{X}_i' \widehat{\boldsymbol{V}}_i^{-1} \mathbf{y}_i \tag{2.19}$$

Because we replaced $\boldsymbol{V}_i$ by its estimate, $\widehat{\boldsymbol{V}}_i$, we say that $\widehat{\boldsymbol{\beta}}$ is the empirical best linear unbiased estimator (EBLUE) of $\boldsymbol{\beta}$.

The variance of $\widehat{\boldsymbol{\beta}}$, $var(\widehat{\boldsymbol{\beta}})$, is a $p \times p$ covariance matrix calculated as follows:

$$var(\widehat{\boldsymbol{\beta}}) = (\sum_i \boldsymbol{X}_i' \widehat{\boldsymbol{V}}_i^{-1} \boldsymbol{X}_i)^{-1} \tag{2.20}$$

We discuss issues related to the estimates of $var(\widehat{\boldsymbol{\beta}})$ in Subsection 2.4.3, because they apply to both ML and REML estimation.

The ML estimates of $\boldsymbol{\theta}$ are biased because they do not take into account the loss of degrees of freedom that results from estimating the fixed-effect parameters in $\boldsymbol{\beta}$ (see Verbeke & Molenberghs (2000) for a discussion of the bias in ML estimates of $\boldsymbol{\theta}$ in the context of LMMs). An alternative form of the maximum likelihood method known as REML estimation

is frequently used to eliminate the bias in the ML estimates of the covariance parameters. We discuss REML estimation in Subsection 2.4.2.

2.4.2 REML Estimation

REML estimation is an alternative way of estimating the covariance parameters in $\boldsymbol{\theta}$. REML estimation (sometimes called residual maximum likelihood estimation) was introduced in the early 1970s by Patterson & Thompson (1971) as a method of estimating variance components in the context of unbalanced incomplete block designs. Alternative and more general derivations of REML are given by Harville (1977), Cooper & Thompson (1977), and Verbyla (1990).

REML is often preferred to ML estimation, because it produces unbiased estimates of covariance parameters by taking into account the loss of degrees of freedom that results from estimating the fixed effects in $\boldsymbol{\beta}$.

The REML estimates of $\boldsymbol{\theta}$ are based on optimization of the following **REML log-likelihood function**:

$$\begin{aligned}\ell_{REML}(\boldsymbol{\theta}) &= -0.5 \times (n-p) \times \ln(2\pi) - 0.5 \times \sum_i \ln(\det(\boldsymbol{V}_i)) \\ &\quad -0.5 \times \sum_i \mathbf{r}_i' \boldsymbol{V}_i^{-1} \mathbf{r}_i - 0.5 \times \sum_i \ln(\det(\boldsymbol{X}_i' \boldsymbol{V}_i^{-1} \boldsymbol{X}_i)) \qquad (2.21)\end{aligned}$$

In this function, $\mathbf{r}_i$ is defined as in (2.17). Once an estimate, $\widehat{\boldsymbol{V}}_i$, of the $\boldsymbol{V}_i$ matrix has been obtained, REML-based estimates of the fixed-effect parameters, $\widehat{\boldsymbol{\beta}}$, and $var(\widehat{\boldsymbol{\beta}})$ can be computed. In contrast to ML estimation, the REML method does not provide a formula for the estimates. Instead, we use (2.18) and (2.19) from ML estimation to estimate the fixed-effect parameters and their standard errors.

Although we use the same formulas in (2.18) and (2.19) for REML and ML estimation of the fixed-effect parameters, it is important to note that the resulting $\widehat{\boldsymbol{\beta}}$ and corresponding $var(\widehat{\boldsymbol{\beta}})$ from REML and ML estimation are different, because the $\widehat{\boldsymbol{V}}_i$ matrix is different in each case.

2.4.3 REML vs. ML Estimation

In general terms, we use maximum likelihood methods (either REML or ML estimation) to obtain estimates of the covariance parameters in $\boldsymbol{\theta}$ in an LMM. We then obtain estimates of the fixed-effect parameters in $\boldsymbol{\beta}$ using results from GLS. However, ML estimates of the covariance parameters are biased, whereas REML estimates are not.

When used to estimate the covariance parameters in $\boldsymbol{\theta}$, ML and REML estimation are computationally intensive; both involve the optimization of some objective function, which generally requires starting values for the parameter estimates and several subsequent iterations to find the values of the parameters that maximize the likelihood function (iterative methods for optimizing the likelihood function are discussed in Subsection 2.5.1). Statistical software procedures capable of fitting LMMs often provide a choice of either REML or ML as an estimation method, with the default usually being REML. Table 2.4 provides information on the estimation methods available in the software procedures discussed in this book.

Note that the variances of the estimated fixed effects, i.e., the diagonal elements in $var(\widehat{\boldsymbol{\beta}})$ as presented in (2.20), are biased downward in both ML and REML estimation, because they do not take into account the uncertainty introduced by replacing $\boldsymbol{V}_i$ with $\widehat{\boldsymbol{V}}_i$

in (2.15). Consequently, the standard errors of the estimated fixed effects, se($\widehat{\boldsymbol{\beta}}$), are also biased downward. In the case of ML estimation, this bias is compounded by the bias in the estimation of $\boldsymbol{\theta}$ and hence in the elements of $\boldsymbol{V}_i$. To take this bias into account, approximate degrees of freedom are estimated for the t-tests or F-tests that are used for hypothesis tests about the fixed-effect parameters (see Subsection 2.6.3.1). Kenward & Roger (1997) proposed an adjustment to account for the extra variability in using $\widehat{\boldsymbol{V}}_i$ as an estimator of $\boldsymbol{V}_i$, which has now been implemented in several of the software procedures discussed in this book (as the example chapters will demonstrate).

The estimated variances of the estimated fixed-effect parameters contained in $var(\widehat{\boldsymbol{\beta}})$ depend on how close $\widehat{\boldsymbol{V}}_i$ is to the "true" value of $\boldsymbol{V}_i$. To get the best possible estimate of $\boldsymbol{V}_i$ in practice, we often use REML estimation to fit LMMs with different structures for the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices and use model selection tools (discussed in Section 2.6) to find the best estimate for $\boldsymbol{V}_i$. We illustrate the selection of appropriate structures for the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ covariance matrices in detail for the LMMs that we fit in the example chapters.

Although we dealt with estimation in the LMM in this section, a very similar algorithm can be applied to the estimation of fixed effects and covariance parameters in the marginal model specified in Section 2.3.

2.5 Computational Issues

2.5.1 Algorithms for Likelihood Function Optimization

Having defined the ML and REML estimation methods, we briefly introduce the computational algorithms used to carry out the estimation for an LMM.

The key computational difficulty in the analysis of LMMs is estimation of the covariance parameters, using iterative **numerical optimization** of the log-likelihood functions introduced in Subsection 2.4.1.2 for ML estimation and in Subsection 2.4.2 for REML estimation, subject to constraints imposed on the parameters to ensure positive definiteness of the $\boldsymbol{D}$ and $\boldsymbol{R}_i$ matrices. The most common iterative algorithms used for this optimization problem in the context of LMMs are the **expectation-maximization (EM) algorithm**, the **Newton–Raphson (N–R) algorithm** (the preferred method), and the **Fisher scoring algorithm**.

The EM algorithm is often used to maximize complicated likelihood functions or to find good starting values of the parameters to be used in other algorithms (this latter approach is currently used by the procedures in R, Stata, and HLM, as shown in Table 2.4). General descriptions of the EM algorithm, which alternates between expectation (E) and maximization (M) steps, can be found in Dempster et al. (1977) and Laird et al. (1987). For "incomplete" data sets arising from studies with unbalanced designs, the E-step involves, at least conceptually, creation of a "complete" data set based on a hypothetical scenario, in which we assume that data have been obtained from a balanced design and there are no missing observations for the dependent variable. In the context of the LMM, the complete data set is obtained by augmenting observed values of the dependent variable with expected values of the sum of squares and sum of products of the unobserved random effects and residuals. The complete data are obtained using the information available at the current iteration of the algorithm, i.e., the current values of the covariance parameter estimates and the observed values of the dependent variable. Based on the complete data, an objective function called the **complete data log-likelihood function** is constructed and maximized in the M-step, so that the vector of estimated $\boldsymbol{\theta}$ parameters is updated at each iteration. The